

# Majorana Modes in the Machine

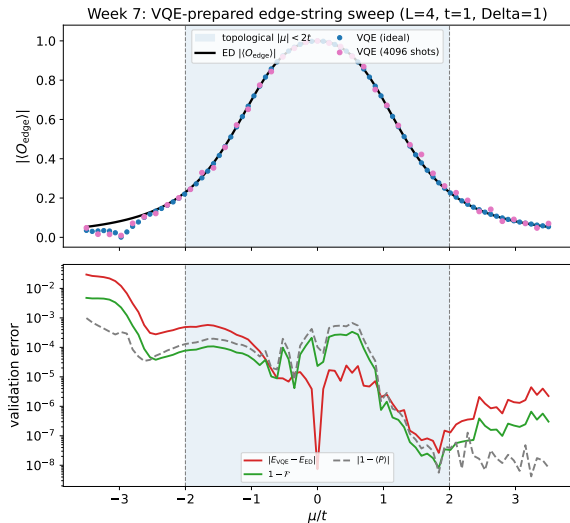
Week 8: NISQ Reality Check – Noise Model and Failure Modes

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# What Week 7 Gives Us



## Noiseless baseline

- Prepared by parity-constrained VQE
- Edge string measured ideally and with shots
- ED validates energy, parity, fidelity, string value

# What Noise Can Break

## State preparation

- VQE ansatz gates no longer implement exactly  $U(\theta)$ .
- Entangling-gate errors reduce edge coherence.
- Relaxation can bias the parity sector.

## Measurement

- Readout flips corrupt the bitstring parity product.
- Shot noise broadens the string estimator.
- A small true signal becomes hard to distinguish from zero.

## Main risk

Noise can flatten the edge-string contrast between  $|\mu| < 2t$  and  $|\mu| > 2t$ , even when the noiseless circuit is correct.

# Noise Channels to Add

| Channel            | Physical meaning                      | Expected effect                               |
|--------------------|---------------------------------------|-----------------------------------------------|
| Depolarizing error | Random Pauli error after gates        | Shrinks correlations and string contrast      |
| Readout error      | Classical bit flip during measurement | Biases $\pm 1$ parity-product estimator       |
| Amplitude damping  | $T_1$ relaxation toward $ 0\rangle$   | Can change particle number and parity balance |
| Phase damping      | $T_2$ dephasing                       | Destroys coherent edge-to-edge correlations   |

# Two Experiments to Keep Separate

## Frozen-parameter noise test

- Use  $\theta^*(\mu)$  from noiseless Week 7.
- Add noise only during circuit execution.
- Measures readout and circuit degradation at fixed preparation.

## Noisy optimization test

- Optimize the VQE cost under noisy estimates.
- Tests whether the optimizer can still find good parameters.
- Harder and more expensive; should come second.

## Engineering rule

First isolate measurement/circuit degradation; only then study optimizer degradation.

# Circuit-Level Noise Protocol: Formalism

**Objective:** Formalize the mapping from the noiseless VQE sweep to the noisy density matrix.

For the Week 7 optimized parameters  $\theta^*(\mu)$ , the ideal pure state is  $\rho_{\text{id}} = |\psi(\theta^*)\rangle\langle\psi(\theta^*)|$ . We define the physical execution via completely positive trace-preserving (CPTP) maps:

$$\rho_{\text{noisy}} = \mathcal{E}_{\text{readout}} \circ \mathcal{E}_{\text{meas-basis}} \circ \mathcal{E}_{\text{gate}}(\rho_{\text{id}}) \quad (1)$$

- **Gate Noise ( $\mathcal{E}_{\text{gate}}$ ):** Local two-qubit depolarizing channels after entangling gates,  $\mathcal{D}_p(\rho) = (1 - p)\rho + \frac{p}{15} \sum_{i=1}^{15} P_i \rho P_i$ , over the non-identity two-qubit Paulis.
- **Basis Rotation ( $\mathcal{E}_{\text{meas-basis}}$ ):** Ideal unitary mapping for  $X_0 Z_1 Z_2 X_3$ .
- **Readout Assignment ( $\mathcal{E}_{\text{readout}}$ ):** Asymmetric bit-flip channel  $p(0|1)$  and  $p(1|0)$  applied to the diagonal elements of the density matrix.

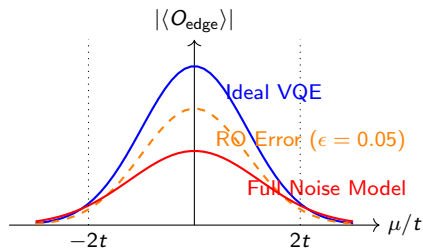
# Frozen-Parameter Test: Readout vs. Depolarizing Channels

**Protocol:** Evaluate  $\text{Tr}(\hat{O}_{\text{edge}}\rho_{\text{noisy}})$  using frozen  $\theta^*(\mu)$  from the ED/ideal VQE baseline.

## Expected Behavior:

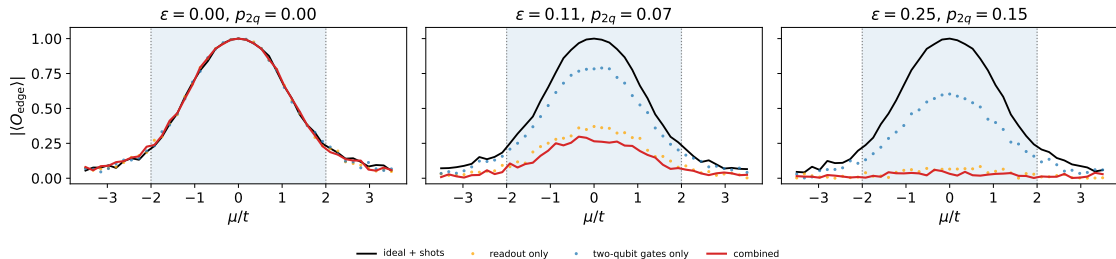
- The ideal ED baseline peaks at  $\langle O_{\text{edge}} \rangle = 1.0$  at  $\mu = 0$ .
- *Readout error alone* scales the curve by  $(1 - 2\epsilon)^4$  for independent symmetric readout flips on the four measured bits.
- *Gate noise* lowers purity,  $\text{Tr}(\rho^2) < 1$ , and suppresses coherent edge correlations, especially near  $\mu_c = \pm 2t$ .

Schematic expectation before simulator data



# Visualising the Noise

Readout and two-qubit depolarizing noise on the edge-string diagnostic



Same Kitaev/Jordan-Wigner Hamiltonian and edge-string operator as Block 3; only the execution noise is varied.



# Phase-Boundary Failure Criterion

We must define mathematically when the topological transition at  $|\mu| = 2t$  becomes indistinguishable from the trivial vacuum due to the combined noise channels.

**Failure Criterion:** The diagnostic fails when the measured contrast between the topological window and the trivial region cannot be resolved from the statistical and hardware noise floor with a  $3\sigma$  confidence interval.

Specifically, failure occurs if the measurable contrast satisfies:

$$\max_{|\mu| < 2t} |\langle \hat{O}_{\text{edge}}(\mu) \rangle_{\text{meas}}| - \max_{|\mu| > 2t} |\langle \hat{O}_{\text{edge}}(\mu) \rangle_{\text{meas}}| \leq \frac{3}{\sqrt{N_{\text{shots}}}} + \delta_{\text{SPAM}} \quad (2)$$

where  $\delta_{\text{SPAM}}$  is the systematic bias inferred from the parity drift observable,  $|1 - \langle \hat{P} \rangle_{\text{noisy}}|$ .

# Next Week

## Next milestone

First produce the noisy frozen-parameter sweep against the Week 7 baseline; then run noisy VQE optimization and compare raw versus mitigated readout.